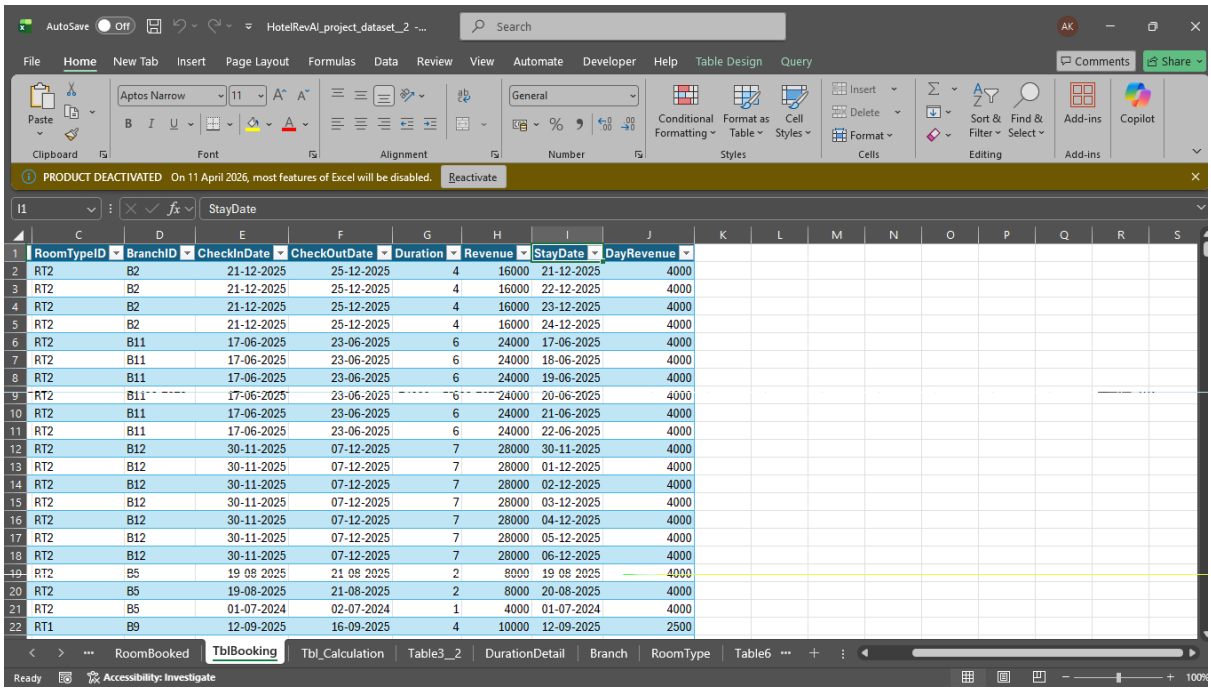
INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

MODULE-3

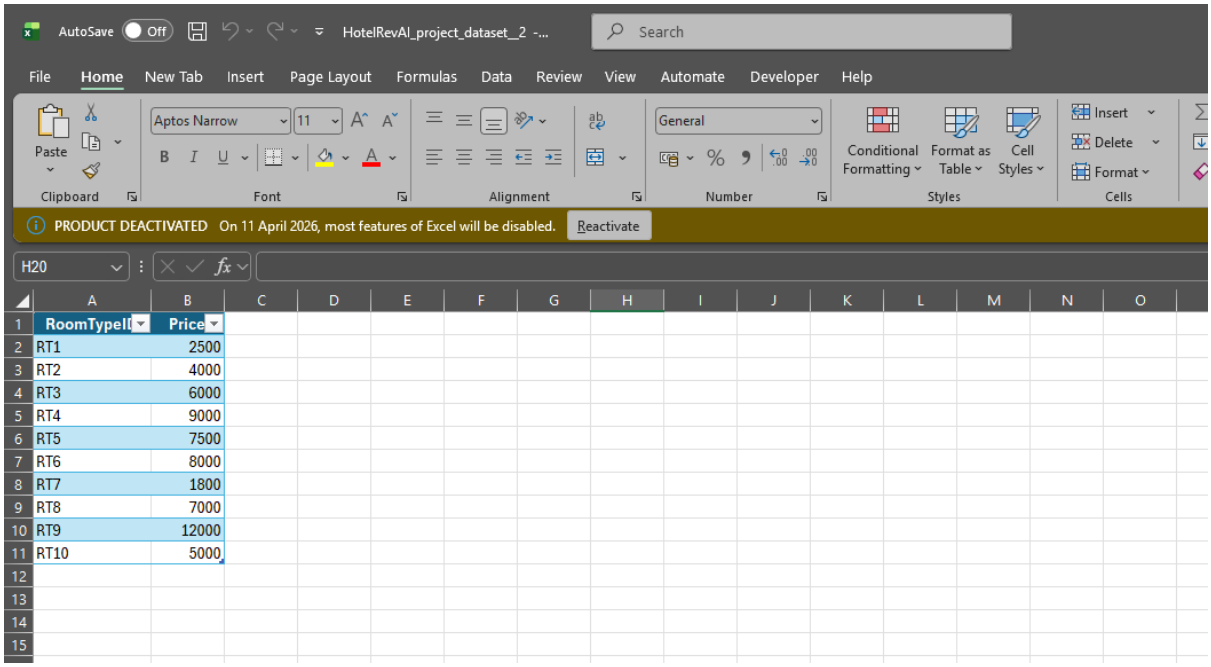
NAME: Abhilash kumar jha

PROJECT NAME: HotelRevAI AI Driven Revenue Analysis for Hotels_Batch-13

1.) Created TblBookings and TblRates by doing merge query in the Power Query Editor.

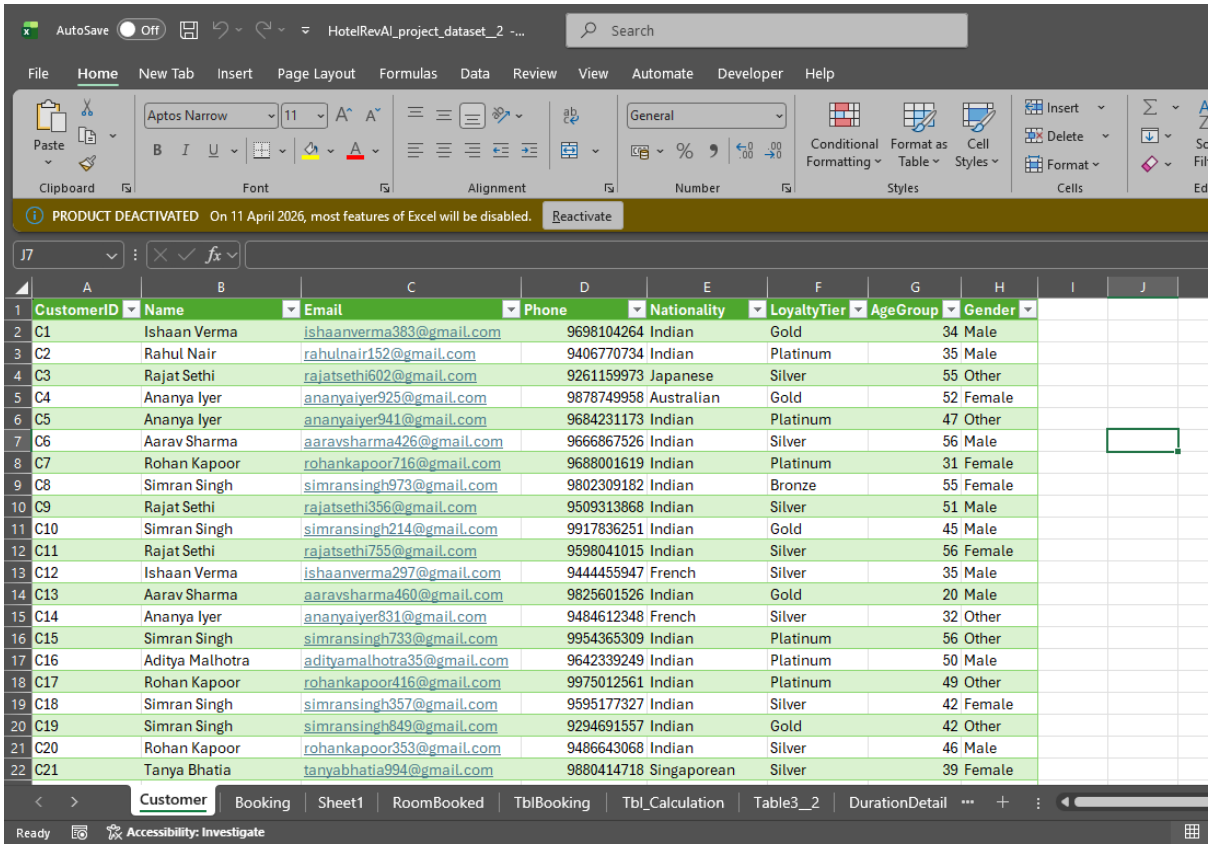


RoomTypeID	BranchID	CheckInDate	CheckOutDate	Duration	Revenue	StayDate	DayRevenue
RT2	B2	21-12-2025	25-12-2025	4	16000	21-12-2025	4000
RT2	B2	21-12-2025	25-12-2025	4	16000	22-12-2025	4000
RT2	B2	21-12-2025	25-12-2025	4	16000	23-12-2025	4000
RT2	B2	21-12-2025	25-12-2025	4	16000	24-12-2025	4000
RT2	B11	17-06-2025	23-06-2025	6	24000	17-06-2025	4000
RT2	B11	17-06-2025	23-06-2025	6	24000	18-06-2025	4000
RT2	B11	17-06-2025	23-06-2025	6	24000	19-06-2025	4000
RT2	B11	17-06-2025	23-06-2025	6	24000	20-06-2025	4000
RT2	B11	17-06-2025	23-06-2025	6	24000	21-06-2025	4000
RT2	B11	17-06-2025	23-06-2025	6	24000	22-06-2025	4000
RT2	B12	30-11-2025	07-12-2025	7	28000	30-11-2025	4000
RT2	B12	30-11-2025	07-12-2025	7	28000	01-12-2025	4000
RT2	B12	30-11-2025	07-12-2025	7	28000	02-12-2025	4000
RT2	B12	30-11-2025	07-12-2025	7	28000	03-12-2025	4000
RT2	B12	30-11-2025	07-12-2025	7	28000	04-12-2025	4000
RT2	B12	30-11-2025	07-12-2025	7	28000	05-12-2025	4000
RT2	B12	30-11-2025	07-12-2025	7	28000	06-12-2025	4000
RT2	B5	19-08-2025	21-08-2025	2	8000	19-08-2025	4000
RT2	B5	19-08-2025	21-08-2025	2	8000	20-08-2025	4000
RT2	B5	01-07-2024	02-07-2024	1	4000	01-07-2024	4000
RT1	B9	12-09-2025	16-09-2025	4	10000	12-09-2025	2500



RoomTypeID	Price
RT1	2500
RT2	4000
RT3	6000
RT4	9000
RT5	7500
RT6	8000
RT7	1800
RT8	7000
RT9	12000
RT10	5000

2.) Added Nationality column in Customer table.



CustomerID	Name	Email	Phone	Nationality	LoyaltyTier	AgeGroup	Gender
C1	Ishaan Verma	ishaanverma383@gmail.com	9698104264	Indian	Gold	34	Male
C2	Rahul Nair	rahulnair152@gmail.com	9406770734	Indian	Platinum	35	Male
C3	Rajat Sethi	rajatsethi602@gmail.com	9261159973	Japanese	Silver	55	Other
C4	Ananya Iyer	ananyaiyer925@gmail.com	9878749958	Australian	Gold	52	Female
C5	Ananya Iyer	ananyaiyer941@gmail.com	9684231173	Indian	Platinum	47	Other
C6	Aarav Sharma	aaravsharma426@gmail.com	9666867526	Indian	Silver	56	Male
C7	Rohan Kapoor	rohankapoor716@gmail.com	9688001619	Indian	Platinum	31	Female
C8	Simran Singh	simransingh973@gmail.com	9802309182	Indian	Bronze	55	Female
C9	Rajat Sethi	rajatsethi356@gmail.com	9509313868	Indian	Silver	51	Male
C10	Simran Singh	simransingh214@gmail.com	9917836251	Indian	Gold	45	Male
C11	Rajat Sethi	rajatsethi755@gmail.com	9598041015	Indian	Silver	56	Female
C12	Ishaan Verma	ishaanverma297@gmail.com	9444455947	French	Silver	35	Male
C13	Aarav Sharma	aaravsharma460@gmail.com	9825601526	Indian	Gold	20	Male
C14	Ananya Iyer	ananyaiyer831@gmail.com	9484612348	French	Silver	32	Other
C15	Simran Singh	simransingh733@gmail.com	9954365309	Indian	Platinum	56	Other
C16	Aditya Malhotra	adityamalhotra35@gmail.com	9642339249	Indian	Platinum	50	Male
C17	Rohan Kapoor	rohankapoor416@gmail.com	9975012561	Indian	Platinum	49	Other
C18	Simran Singh	simransingh357@gmail.com	9595177327	Indian	Silver	42	Female
C19	Simran Singh	simransingh849@gmail.com	9294691557	Indian	Gold	42	Other
C20	Rohan Kapoor	rohankapoor353@gmail.com	9486643068	Indian	Silver	46	Male
C21	Tanya Bhatia	tanyabhatia994@gmail.com	9880414718	Singaporean	Silver	39	Female

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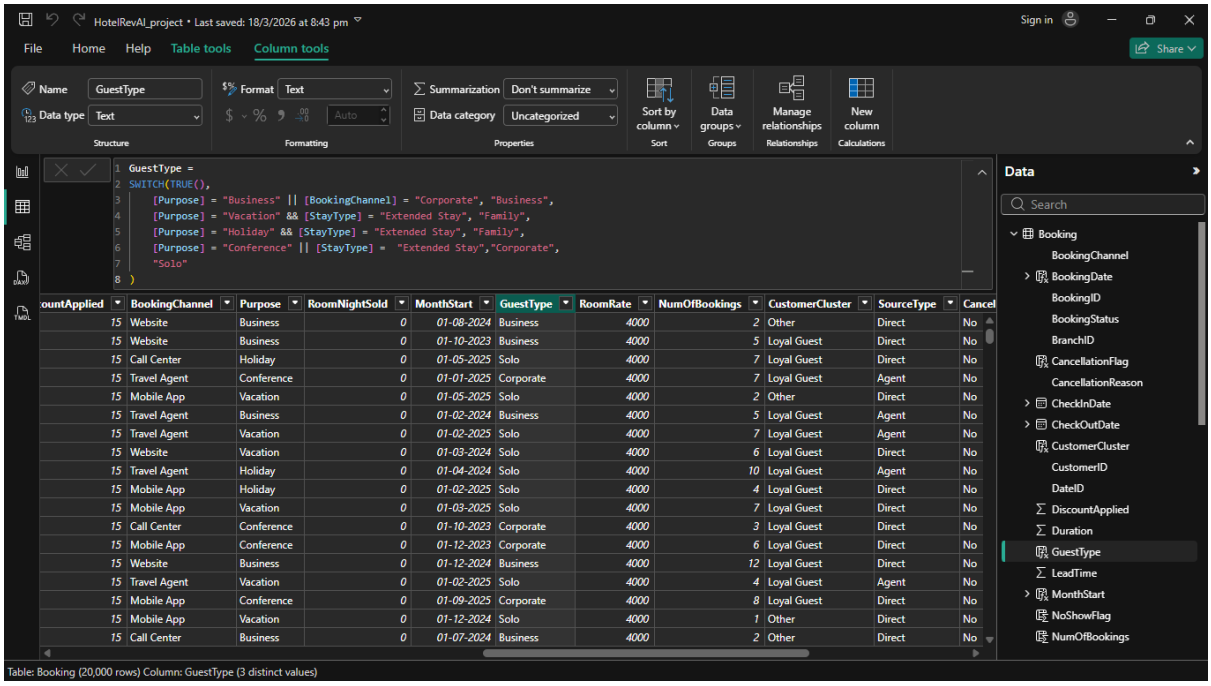
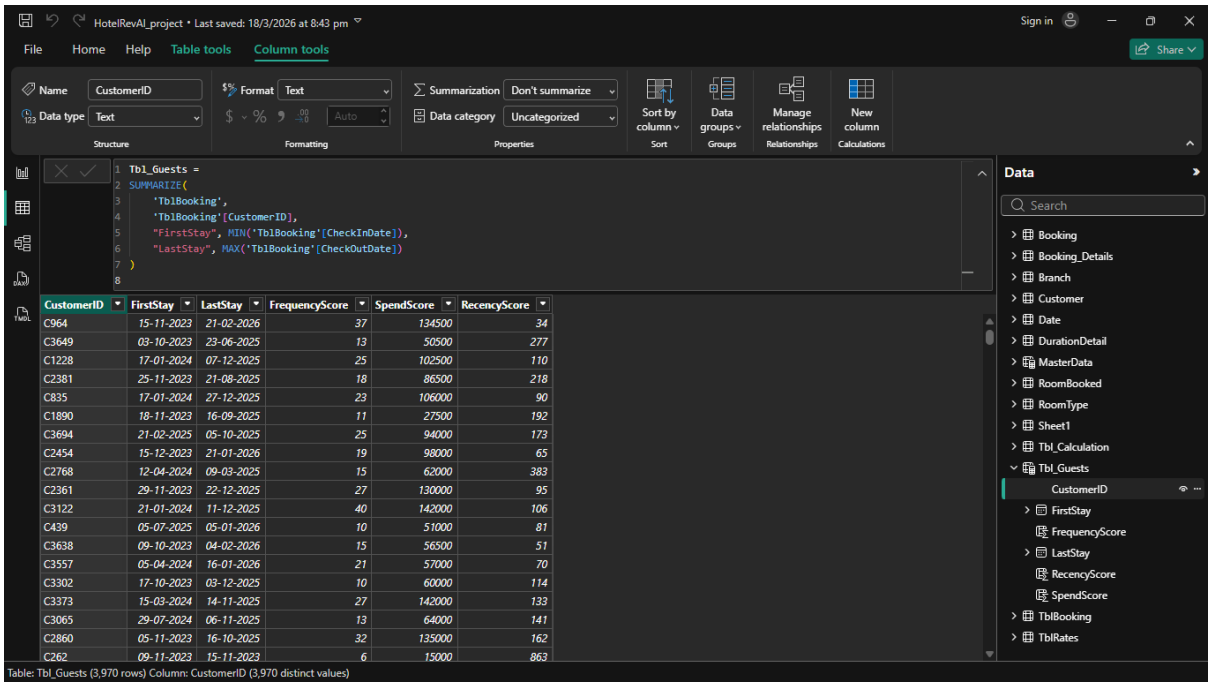
3.) In Power BI, refresh and load the remaining data in it and then we have added SourceType, GuestType, RoomRate, NumOfBookings, CustomerCluster columns in bookings table.




```
SourceType =
SWITCH(
   'Bookings'[BookingChannel],
   "Call Center", "Direct",
   "Mobile App", "Direct",
   "Website", "Direct",
   "Travel Agent", "OTA",
   "Other"
)

GuestType =
SWITCH(TRUE(),
   [Purpose] = "Business" || [BookingChannel] = "Corporate", "Business",
   [Purpose] = "Vacation" && [StayType] = "Extended Stay", "Family",
   [Purpose] = "Holiday" && [StayType] = "Extended Stay", "Family",
   [Purpose] = "Conference" || [StayType] =  "Extended Stay","Corporate",
   "Solo"
)

RoomRate = RELATED(Room[Price])

NumOfBookings =
CALCULATE(
   COUNT(Bookings[BookingID]),
```

```
    ALLEXCEPT(Bookings, Bookings[CustomerID])
)

CustomerCluster =
SWITCH(TRUE(),
    [NumOfBookings] = 1 && [Revenue] < 5000, "First-Timer",
    [NumOfBookings] >= 3 && [Revenue] < 15000, "Loyal Guest",
    [Revenue] >= 15000, "High Spender",
    "Other"
)
```

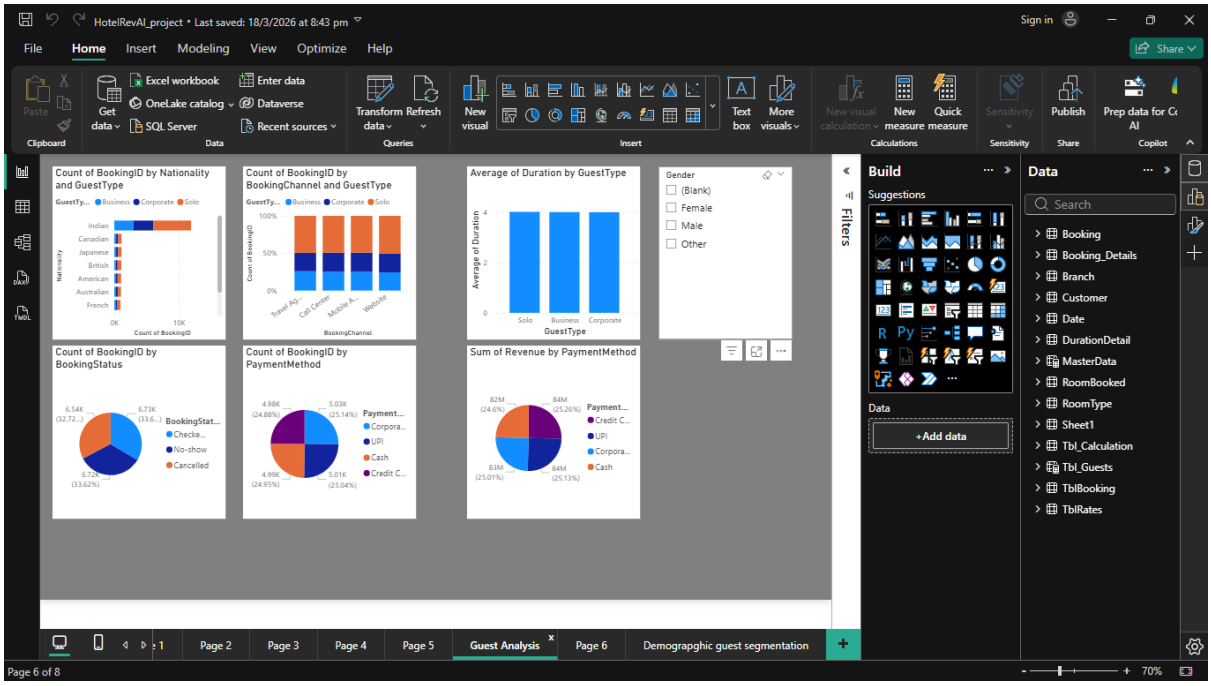
4.) Then we have created a table named as TblGuests in Power BI and added columns named as FrequencyScore, SpendScore, RecencyScore.



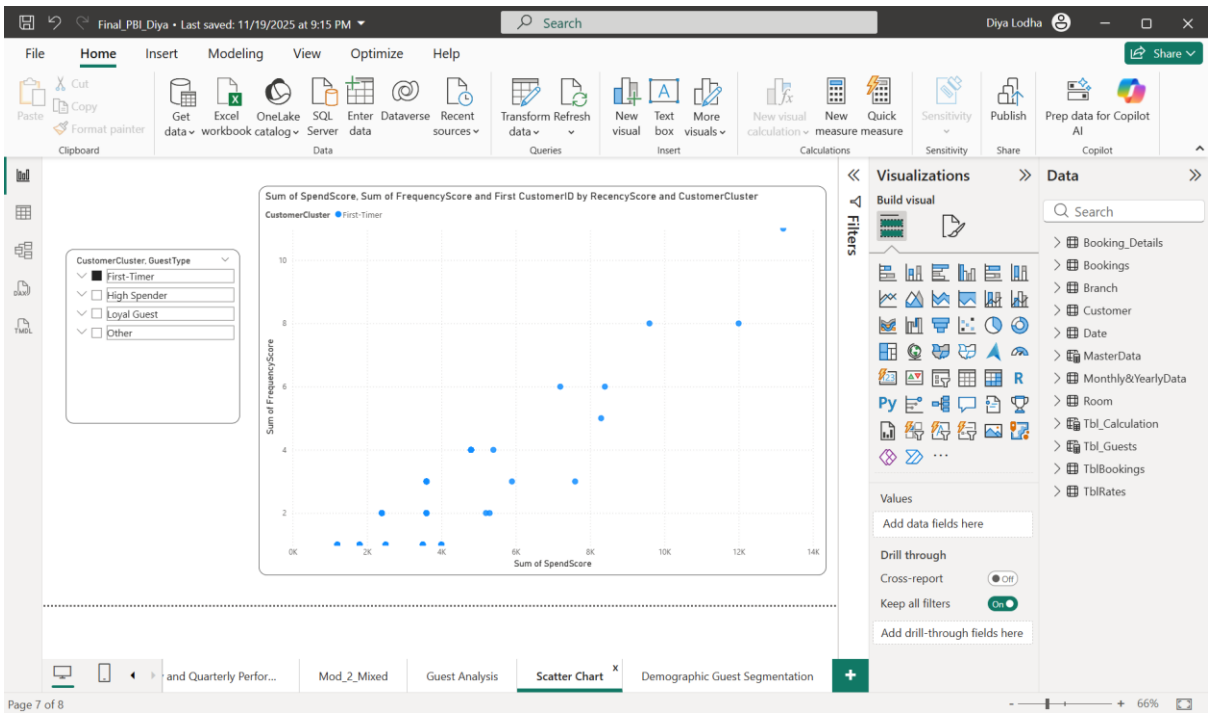
CustomerID	FirstStay	LastStay	FrequencyScore	SpendScore	RecencyScore
C964	15-11-2023	21-02-2026	37	134500	34
C3649	03-10-2023	23-06-2025	13	50500	277
C1228	17-01-2024	07-12-2025	25	102500	110
C2381	25-11-2023	21-08-2025	18	86500	218
C835	17-01-2024	27-12-2025	23	106000	90
C1890	18-11-2023	16-09-2025	11	27500	192
C3694	21-02-2025	05-10-2025	25	94000	173
C2454	15-12-2023	21-01-2026	19	98000	65
C2768	12-04-2024	09-03-2025	15	62000	383
C2361	29-11-2023	22-12-2025	27	130000	95
C3122	21-01-2024	11-12-2025	40	142000	106
C439	05-07-2025	05-01-2026	10	51000	81
C3638	09-10-2023	04-02-2026	15	56500	51
C3557	05-04-2024	16-01-2026	21	57000	70
C3302	17-10-2023	03-12-2025	10	60000	114
C3373	15-03-2024	14-11-2025	27	142000	133
C3065	29-07-2024	06-11-2025	13	64000	141
C2860	05-11-2023	16-10-2025	32	135000	162
C262	09-11-2023	15-11-2023	6	15000	863

VISUALISATION

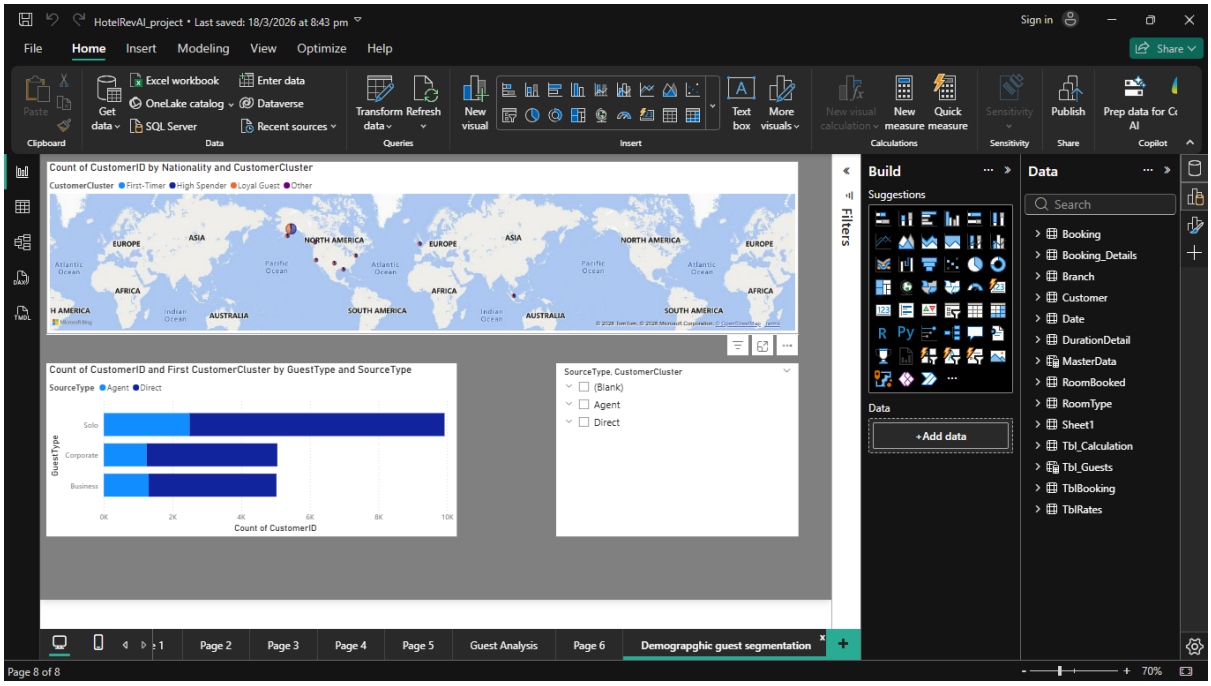
Guest Analysis:



Scatter Chart (Based on sum of SpendScore, sum of FrequencyScore and CustomerID by RecencyScore and CustomerCluster):



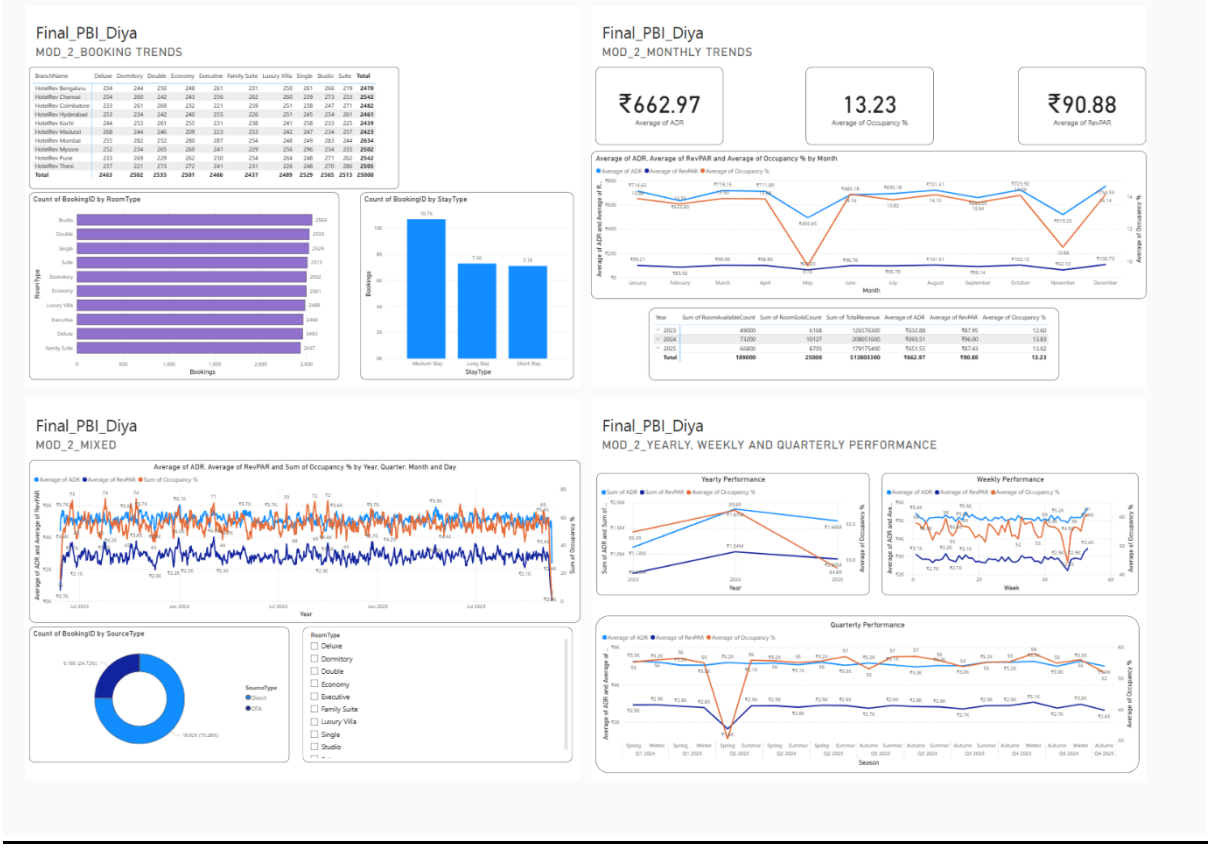
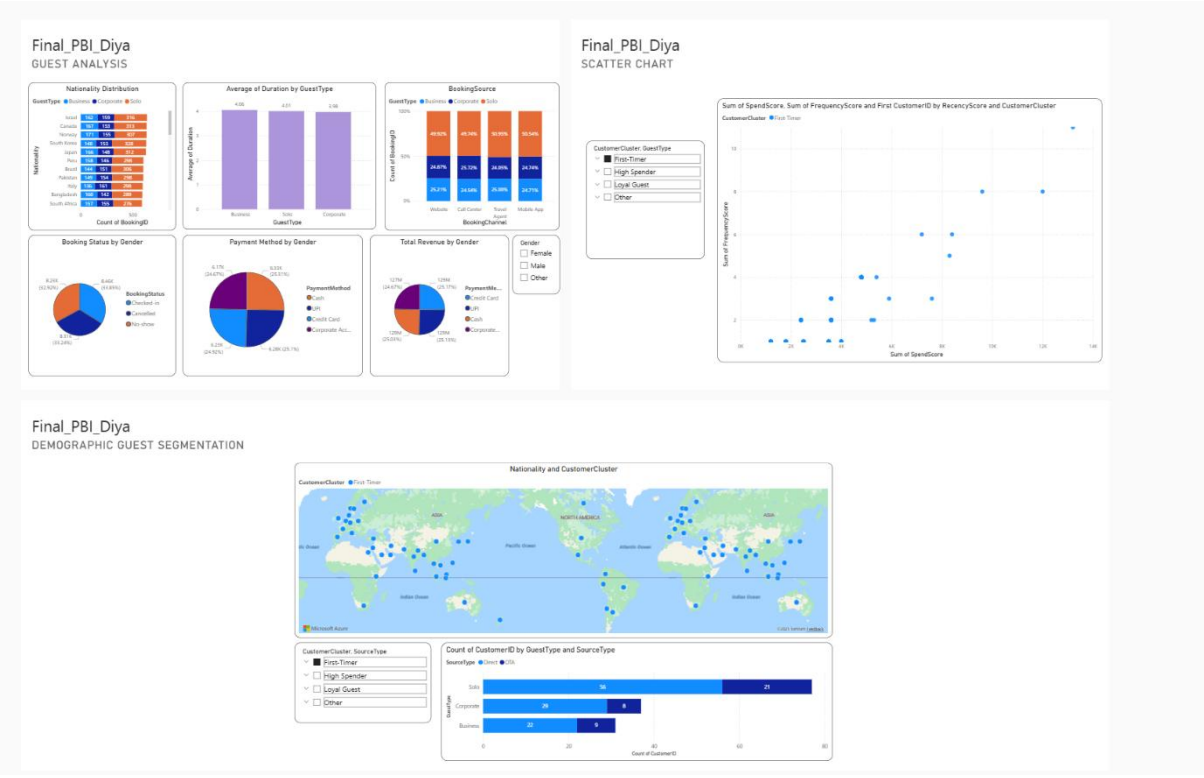
Demographic Guest Segmentation (Based on Nationality , CustomerCluster, SourceType, GuestType):



DASHBOARDS

Module 1 Dashboard: Overview

Module 2 Dashboard: Trends with respect to ADR, Occupancy% and RevPAR



Module 3 Dashboard : Guest Analysis